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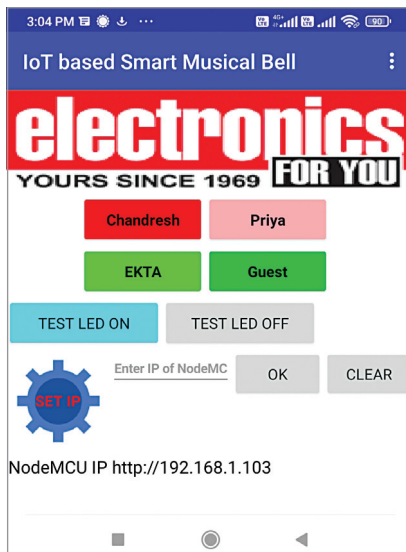


Fig. 3: Music bell Android application

- Operating temperature range of $-40^{\circ}\text{C} \sim 125^{\circ}\text{C}$

The pin diagram of NodeMCU is shown in Fig. 2.

When a person is near the IR sensor, it produces logic 1 at the output that is connected to GPIO4 (pin D2) of NodeMCU. Program is written in such a way that when D2 pin is high, the guest tune is played. All family members will have the Android application installed in their smartphones and use the Wi-Fi at home. So, when any member rings bell the corresponding tune will play and identify the member.

IR sensor is for guests and strangers while smartphone application is for family members. If a family member is not carrying his smartphone or doesn't use it, he/she will have to ring the bell like a guest and guest tune will play whenever finger comes near the infrared sensor. It will also thus work as a touch-free bell in the present Covid-19 situations and help prevent spreading of the virus.

Software

NodeMCU board can be programmed with Arduino IDE that can be installed, along with NodeMCU support, using Board Manager. This installation

PARTS LIST

Semiconductors:

- Board1 - NodeMCU ESP8266-12E
- T1 - BC547 NPN transistor
- IR module - 3-pin infrared sensor module

Resistor (1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-ohm

Miscellaneous:

- LS1 - 8-ohm, 0.5W loudspeaker
- Jumper wires
- 5V DC power supply board
- Android mobile with music bell app

process is not being included here as it has already been published in EFY and can be seen at <https://www.electronicsforu.com/electronics-projects/authentication-device-control-circuit>

After writing the source code *MusicalBell.ino* in Arduino IDE, press upload button to send code to NodeMCU board. Open the serial monitor by clicking on serial monitor option from Tools menu, select the baud as 19200 and press RST button on NodeMCU. NodeMCU will connect with the Wi-Fi network. Note down IP address because you will need it for the Android app later.

Construction and testing

Wire the circuit on a breadboard or general-purpose PCB and upload the code to NodeMCU. Install Android app *Musical Bell.apk* in your smartphone, open the app and enter the IP address of NodeMCU in your mobile application as shown in Fig. 3. Once connected successfully, you can press any of the buttons (Chandresh, Priya, Ekta, or Guest) to ring the doorbell. Each button produces a unique tone.

You can change the name of each button as per your requirement in the Android source code. You can also install the IR sensor module near the existing doorbell switch. Cover it with suitable enclosure so that it is not disturbed by any source of light including sunlight. The front side of the sensor should be facing outside such that when you put your hand near the sensor, the doorbell rings. This IR sensor is for the guests or those who do not have the mobile app in their smartphones. **EFY**

EFY Note
The source code of this project is available for free download at source.efymag.com



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